



Lab 03: Service Accounts and Roles - Fundamentals

Cloud Security Fundamentals | GSP199

Overview

Created and managed service accounts, granted IAM roles to them, and used a service account attached to a VM to query BigQuery public datasets using Python client libraries.

Key Concepts

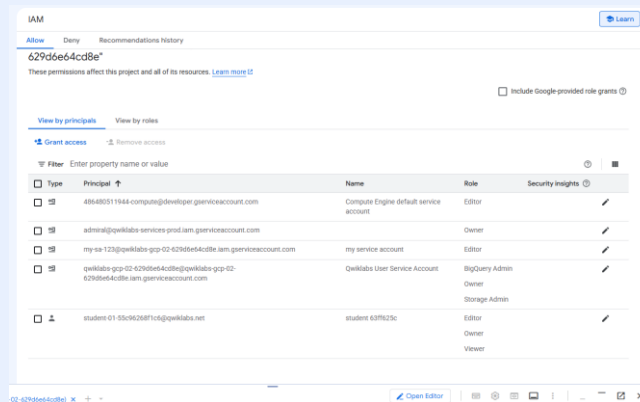
Type	Description
User-managed SA	Created by you for your applications and VMs
Google-managed SA	Created and managed by Google for internal processes
Compute Engine SA	Default SA for Compute Engine: PROJECT_NUMBER-compute@developer.gserviceaccount.com

Objectives

- Create and manage service accounts
- Grant IAM roles to a service account
- Create a VM with a service account attached
- Use Python client libraries to query BigQuery

Task 1 - Create and Manage Service Accounts

```
gcloud iam service-accounts create my-sa-123 --display-name "my service account"
gcloud projects add-iam-policy-binding $DEVSHHELL_PROJECT_ID --member
serviceAccount:my-sa-123@$DEVSHHELL_PROJECT_ID.iam.gserviceaccount.com --role
roles/editor
```



Task 2 - Use Service Account with BigQuery

Created bigquery-qwiklab service account with BigQuery Data Viewer and BigQuery User roles. Created a VM with the service account attached and used Python to query a public BigQuery dataset.

- Service account: bigquery-qwiklab
- Roles: BigQuery Data Viewer and BigQuery User
- VM name: bigquery-instance, machine type: e2-medium

```
sudo apt install python3 python3-pip python3.11-venv -y
pip3 install google-cloud-bigquery pyarrow pandas db-dtypes
python3 query.py
```

```
project='qwiklabs-gcp-02-629d6e64cd8e',
credentials=credentials)
print(client.query(query).to_dataframe())

(myvenv) student-01-55c96268f1c6@bigquery-instance:~$ python3 query.py
/home/student-01-55c96268f1c6/myvenv/lib/python3.11/site-packages/google/cloud/bigquery/table.py:2086: UserWarning: Bi
gQuery Storage module not found, fetch data with the REST endpoint instead.
  warnings.warn(
  year  num_babies
0  2007      4324008
1  2003      4096092
2  2002      4027376
3  2008      4255156
4  2001      4031531
5  2005      4145619
6  2006      4273225
7  2004      4118907
(myvenv) student-01-55c96268f1c6@bigquery-instance:~$
```

What I Learned

- Service accounts grant permissions to applications and VMs not end users
- VMs can be assigned a service account to access GCP APIs securely
- Python client libraries use the VM service account automatically via compute engine.
- BigQuery can be queried programmatically using the google-cloud-bigquery library
- Least privilege means only granting BigQuery roles needed for the specific query

Resources

GCP Service Accounts Docs: <https://cloud.google.com/iam/docs/service-accounts>

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